LAADAN-AC: A Lagrangian Admissibility-Aware Actor-Critic

From ICU-Sepsis Benchmarking toward Safe Adaptive Robot Control

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Research Note

Abstract

This research note presents LAADAN-AC, a Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic proposal developed from an Intelligent Adaptive Systems coursework project. The immediate benchmark setting is ICU-Sepsis, where the agent learns an offline policy over a fixed tabular Markov decision process. The wider research aim is to investigate whether the same safety structure can later support adaptive robot-control systems, especially in social robotics where unsafe or unsupported actions must be avoided. LAADAN-AC combines actor-critic learning with admissibility masking, twin reward critics, a cost critic, Lagrangian cost control, expert-policy regularisation and smoothness-aware action penalties. The constrained formulation is grounded in constrained Markov decision processes and constrained reinforcement learning, while the admissibility mask, expert guidance and smoothness term are project-specific design choices. The future direction is to connect this safety-aware policy layer with JEPA-inspired latent world models for safer adaptive robot behaviour.

Keywords: offline reinforcement learning; constrained Markov decision process; actor-critic learning; Lagrangian safety; admissibility masking; social robotics; world models; JEPA.

1. Introduction

Adaptive agents should not only optimise reward. In safety-critical environments, they must also avoid actions that are unsafe, unsupported or behaviourally implausible. This is the central motivation behind LAADAN-AC. The model was first implemented on the ICU-Sepsis benchmark, where the agent chooses among 25 discretised treatment actions from a fixed state representation. However, the long-term research vision is broader: to explore whether the same constrained actor-critic structure can become a safer adaptive-control framework for real-world and social robotics.

The design follows the constrained Markov decision process view, where a policy maximises expected return while satisfying a cost constraint (Altman, 1999). In reinforcement learning, this principle also appears in constrained policy optimisation, where policy improvement is coupled with explicit safety or cost constraints (Achiam et al., 2017). LAADAN-AC adopts this general constrained-learning motivation, but combines it with project-specific mechanisms for admissibility-aware action selection and expert-guided policy shaping.

2. Model Overview

LAADAN-AC contains four main components:

- an **actor** that maps a state s to action logits $z_\theta(s, a)$;
- **twin reward critics** Q_1 and Q_2, used to estimate long-term reward value;
- a **cost critic** C_ψ, used to estimate unsafe or inadmissible action cost;
- an **admissibility mask** $m(s, a)$, used to block actions that are not allowed in a given state.

For state s and action a, the actor first produces logits $z_\theta(s,a)$. LAADAN-AC then applies an admissibility mask:

$$\pi_\theta(a|s) = \frac{\exp(z_\theta(s,a))m(s,a)}{\sum_{a'} \exp(z_\theta(s,a'))m(s,a')}.$$

Here, $m(s,a) = 1$ means that action a is admissible in state s, while $m(s,a) = 0$ means that the action is blocked. This masking mechanism is a project-specific LAADAN-AC design choice. It provides the direct action-level safety filter used by the final policy.

3. Notation

Symbol	Meaning
s	Current state.
a	Candidate action.
$\pi_\theta(a\|s)$	Actor policy parameterised by θ.
$z_\theta(s,a)$	Actor logit for action a in state s.
$m(s,a)$	Admissibility mask; 1 for allowed actions, 0 for blocked actions.
Q_1, Q_2	Twin reward critics.
$Q(s,a)$	Conservative reward estimate, defined as $\min(Q_1, Q_2)(s,a)$.
$C_\psi(s,a)$	Cost critic estimating unsafe or inadmissible cost.
π_E	Expert-safe policy used for regularisation.
$S(s,a)$	Smoothness or action-deviation cost used in this project.
λ	Lagrange multiplier controlling cost pressure.
d	Allowed cost budget.
α, β, μ	Weights for entropy, expert guidance and smoothness.

4. Constrained Objective

The safety problem is written as a constrained optimisation problem:

$$\max_{\pi_\theta} J_R(\pi_\theta) \quad \text{subject to} \quad J_C(\pi_\theta) \le d.$$

Here, $J_R(\pi_\theta)$ is the expected return and $J_C(\pi_\theta)$ is the expected unsafe cost. In the current implementation, $d = 0$, so the policy is encouraged toward zero unsafe cost.

Following constrained reinforcement learning, the constraint can be relaxed using a non-negative Lagrange multiplier λ (Altman, 1999; Achiam et al., 2017):

$$\mathcal{L}(\pi_\theta, \lambda) = J_R(\pi_\theta) - \lambda \left(J_C(\pi_\theta) - d\right).$$

In the LAADAN-AC implementation, this idea appears in the actor objective as an adaptive cost penalty.

5. Actor Objective

The actor minimises the following proposed LAADAN-AC loss:

$$\mathcal{L}_{actor} = \mathbb{E}_s \left[-\mathbb{E}_{a\sim\pi_\theta}[Q(s,a)] - \alpha\mathcal{H}(\pi_\theta) + \lambda\mathbb{E}_{a\sim\pi_\theta}[C_\psi(s,a)] + \beta D_{KL}(\pi_E \| \pi_\theta) + \mu\mathbb{E}_{a\sim\pi_\theta}[S(s,a)] \right].$$

The first term maximises reward value. The entropy term encourages a less brittle policy and is motivated by maximum-entropy actor-critic learning (Haarnoja et al., 2018). The cost term penalises unsafe predicted cost. The expert KL term keeps the policy close to the expert-safe policy. The smoothness term discourages unusually rough or extreme action choices. The expert KL and smoothness terms are project-specific LAADAN-AC additions.

The Lagrange multiplier is updated by projected ascent:

$$\lambda \leftarrow \max\left(0, \lambda + \eta_\lambda(\bar{C} - d)\right),$$

where η_λ is the Lagrange learning rate and $\bar{C}$ is the current mean expected cost. If the expected cost exceeds the budget, λ increases, making unsafe actions more expensive in later updates. This is the mathematical reason LAADAN-AC is described as Lagrangian.

6. Critic Learning

The reward and cost critics are fitted to finite-horizon Bellman targets computed from the benchmark transition and reward functions. The reward critics estimate long-term return, while the cost critic estimates long-term unsafe cost. A conservative critic regulariser is also used to discourage unsupported value inflation, following the motivation of Conservative Q-Learning for offline reinforcement learning (Kumar et al., 2020).

The important distinction is that the hard zero-inadmissibility behaviour mainly comes from the admissibility mask, while the Lagrangian term adds adaptive pressure during learning. LAADAN-AC is not claiming that the Lagrangian term alone guarantees safety. Instead, safety is encouraged through a combination of masking, cost estimation, expert guidance and constrained optimisation.

7. Future Research Vision

The current ICU-Sepsis implementation is a benchmark prototype, not a clinical deployment system. The future research direction is to adapt LAADAN-AC beyond a tabular MDP into safer adaptive robot control. This is motivated by the idea that autonomous agents should learn predictive world models, reason in latent representation space and use those predictions to guide action selection (LeCun, 2022).

A future JEPA-inspired LAADAN-AC would replace the fixed benchmark state with a learned latent world model. In a social robotics setting, the robot would use this world model to predict likely consequences of actions before acting. The LAADAN-AC policy layer would then apply admissibility constraints, Lagrangian cost control, expert guidance and smoothness penalties to select safer actions.

Recent JEPA-based work such as LeWorldModel suggests that joint-embedding predictive architectures can learn compact latent world models from visual inputs for prediction and planning (Maes et al., 2026). This makes LAADAN-AC a potential bridge between safe offline reinforcement learning and world-model-based adaptive robotics.

8. Limitations

This proposal should be read as an early-stage research note. The current implementation is evaluated on a benchmark MDP and should not be interpreted as a real clinical or robotics deployment system. Further work would need stronger ablations, broader benchmarks, external validation and direct testing in simulated robotics environments before any real-world claims could be made.

References

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